AW Notes

Laurin Seeholzer

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Contents

1 Connectivness of Graphs **3**

2 Eulertours **6**

3 Hamiltonian Cycles **7**
3.1 Travelling Salesman Problem (TSP) 7
3.1.1 Metric TSP 8

4 Matchings **9**

5 Coloring **10**

6 Intro: Probability **11**
6.1 Conditional Probability 12
6.2 Independence 14

7 Random Variables **15**
7.1 Expectation 15
7.2 Variance 17

8 Distributions **18**

1 Connectivness of Graphs

Definition 1.1. *Let $G = (V, E)$ be a graph, $u, v \in V, u \neq v$. u and v are called* ***connected*** *if there is a $u - v$-path in G.*

Definition 1.2. *A graph $G = (V, E)$ is called* ***connected*** *if $\forall u, v \in V, u \neq v$ a $u - v$-path in G exists.*

Definition 1.3. *Let $G = (V, E)$ be a graph, $u, v \in V, u \neq v$. u and v are called* ***k-connected*** *if $|V| \geq k + 1$ and*

$$\forall X \subseteq V \setminus \{u, v\} \text{ with } |X| < k, u \text{ and } v \text{ are connected in } G[V \setminus X]$$

Definition 1.4. *A graph $G = (V, E)$ is called* ***k-connected*** *if $|V| > k$ and*

$$\forall X \subseteq V \text{ with } |X| < k, G[V \setminus X] \text{ is connected}$$

Definition 1.5. *Let $G = (V, E)$ be a graph, $u, v \in V$ and $X \subseteq V \setminus \{u, v\}$. X is called a $u - v$-****separator*** *if u and v are in different connected components of $G[V \setminus X]$.*

More generally one can say that if a Graph G contians a vertex v stuch that $deg(v) < k$, then G can not be k-connected.

Theorem 1.6. ***Menger's Theorem (Vertex Version)****:*

Let $G = (V, E)$ be a graph and $u, v \in V$. The minimum size of a $u - v$-separator is equal to the maximum number of internally vertex-disjoint $u - v$-paths.

A graph G is k-connected iff. $\forall u, v \in V, u \neq v$ there are k internally vertex-disjoint $u - v$-paths.

Definition 1.7. *A graph $G = (V, E)$ is called* ***k-edge-connected*** *if*

$$\forall X \subseteq E \text{ with } |X| < k, (V, E \setminus X) \text{ is connected}$$

Theorem 1.8. ***Menger's Theorem (Edge Version):***

A graph G is k-edge-connected iff. $\forall u, v \in V, u \neq v$ there are k edge-disjoint $u - v$-paths.

Theorem 1.9. *It always holds that:*

$$\textit{Vertex Connectivity} \leq \textit{Edge Connectivity} \leq \textit{Minimum Degree } \delta(G)$$

Lets look at the special case where $k = 1$

Definition 1.10. *Let $G = (V, E)$ be a connected graph.*

- *A vertex $v \in V$ is called a **cut vertex** (Artikulationsknoten) iff. $G[V \setminus \{v\}]$ is disconnected.*
- *An edge $e \in E$ is called a **bridge** (Brücke) iff. $(V, E \setminus \{e\})$ is disconnected.*

Lemma 1.11. *If $e = \{x, y\} \in E$ is a bridge, then $deg(x) = 1$ or x is a cut vertex (analogously for y).*

Definition 1.12. ***Blocks:***

We define an equivalence relation $\sim$ on E by:

$$e \sim f \iff e = f \vee \exists \text{ cycle containing } e \text{ and } f$$

*The equivalence classes are called **blocks**.*

Note that two blocks can share at most one vertex, more over this vertex has to be a cut vertex.

Definition 1.13. ***Block Graph:***

The block graph of a connected graph G is a bipartite graph $T = (A \cup B, E_T)$ where:

- $A = \{\textit{cut vertices of } G\}$
- $B = \{\textit{blocks of } G\}$
- *Edge $\{a, b\} \in E_T \iff a$ is incident to an edge in block b.*

Theorem 1.14. *If G is connected, the block graph of G is a tree.*

Algorithm 1.15

```

#Depth-First Search (DFS) \& Low Values
#For efficient computation of cut vertices and bridges in time $O(|V|+|E|)$.

low[v] := #the smallest DFS number reachable from $v$ by a directed path consisting of tree edges and at most one back edge.

#It holds that:
low[v] = min{ dfs[v], min_{(v,w)  back edge} dfs[w], min_{(v,w) tree edge} low[w] }

#A vertex v is a cut vertex iff:

v != root and there exists a child u with low[u] >= df s[v].
v = root and has >= 2 children in the DFS tree.

A tree edge (v,w) is a bridge iff low[w] >= df s[v].
```

2 Eulertours

Definition 2.1. *An **Eulertour** is a closed walk in a graph G that contains every edge exactly once.*

Theorem 2.2. *A connected graph $G = (V, E)$ contains an Eulertour iff the degree of every vertex is even.*

We can find an Eulertour in time $O(|E|)$. Using an algorithm illustrated by a Runner and a Turtle. The Runner is set lose and traverses the graph until he reaches a vertex he has already visited (we found a cycle). Then the Turtle is set lose and traverses the graph along that path until it reaches a vertex that has unvisited neighbours, from where on the Runner is set lose again (we expand the cycle by findin a new one adjacent to it). This process expands the tour by a cycle at a time until all edges have been traversed.

3 Hamiltonian Cycles

Definition 3.1. *A **Hamiltonian Cycle** is a cycle in $G = (V, E)$ that contains every vertex exactly once.*

Theorem 3.2. *Let $m, n \geq 2$. An $n \times m$ grid contains a Hamiltonian cycle iff $n \cdot m$ is even.*

Theorem 3.3. *A bipartite graph $G = (A \cup B, E)$ with $|A| \neq |B|$ cannot contain a Hamiltonian cycle.*

Theorem 3.4. ***Dirac's Theorem (1952):***

Every graph $G = (V, E)$ with $|V| \geq 3$ and minimum degree $\delta(G) \geq |V|/2$ contains a Hamiltonian cycle.

Algorithm 3.5

```

#Dynamic Programming for Hamiltonian Cycle
#Time complexity O(n^2 * 2^n).
#Define P[S, x] = 1 if there is a 1-x path containing exactly the
    vertices in S.

P[S, x] = max { P[S \ {x}, y] | y in N(x) intersect S, y != 1 }
```

3.1 Travelling Salesman Problem (TSP)

The travelling salesman problem is the problem of finding a Hamiltonian cycle in a complete graph K_n with minimal weight. In other words, the shortes roundtrip/roundtour.

Theorem 3.6. *The travelling salesman problem is NP-Complete.*

This is easy to prove by reduction from Hamiltonian Cycle. Assume we have a Graph G and want to check if it is Hamiltionian. Now lets extend the graph to be complete and set the weight w_e of every edge $e = (u, v) \in G$ to 0 and the weight of all the newly added edges to 1. Now we can check if the TSP has a solution with weight 0. If it does, then G has a Hamiltonian cycle, otherwise it does not. Therfore TSP has to be NP-Complete as Hamiltionian decision is to.

3.1.1 Metric TSP

Assume we have a graph G for which the triangle inequality holds, i.e.:

$$w(u,v) \leq w(u,w) + w(w,v) \quad \forall u,v,w \in V$$

For this problem there is a 2 and even a 1.5 approximation algorithm. **Algorithm 3.7**

```

#2-Approximation for Metric TSP
#This Hamiltonian cycle has length at most 2 * |E(T)| = 2 * (|V|-1)
    .

determine a minimal-spanning tree T
double all the edges of T to get an Eulerian multigraph T'
go along that Eulertour and skip every vertex that has already been
     visited
yield the resulting Hamiltonian cycle
```

4 Matchings

Definition 4.1. *An edge set $M \subseteq E$ is called a* ***matching*** *in $G = (V, E)$ if no vertex is incident to more than one edge from M.*

- *A vertex v is* ***covered*** *by M if it is incident to an edge in M.*
- *A matching is* ***perfect*** *if every vertex is covered ($|M| = |V|/2$).*
- *A matching is* ***maximal*** *if it cannot be extended by adding edges.*
- *A matching is* ***maximum*** *if there is no matching with more edges.*

Algorithm 4.2

```

#Greedy Algorithm
#This leads to a maximal matching M.
#The size of M is at least half the size of a maximum matching.

M = empty set
while E is not empty:
    choose an arbitrary edge e from E
    add e to M
    remove all edges incident to the endpoints of e from E
```

Theorem 4.3. ***Hall's Marriage Theorem:***

A bipartite graph $G = (A \cup B, E)$ contains a matching of cardinality $|M| = |A|$ iff:

$$\forall X \subseteq A : |X| \leq |N(X)|$$

Theorem 4.4. ***Frobenius Corollary:***

Every k-regular bipartite graph contains a perfect matching.

Definition 4.5. *An M-**augmenting path** is a path that alternates between edges in M and not in M, starting and ending in uncovered vertices.*

Theorem 4.6. ***Berge's Theorem (1957):***

A matching M is maximum iff there is no M-augmenting path.

5 Coloring

Definition 5.1. *A* k***-coloring*** *of* $G = (V, E)$ *is a mapping* $c : V \to [k]$ *such that* $c(u) \neq c(v)$ *for all* $\{u, v\} \in E$.

The ***chromatic number*** $\chi(G)$ *is the minimum number of colors.*

$\chi(G) \leq 2 \iff G$ *is bipartite.*

Theorem 5.2. *For* $k \geq 3$*, the problem "Is* $\chi(G) \leq k$*?" is NP-complete.*

Algorithm 5.3

```

#Greedy Coloring
#Requires at most maximum degree + 1 colors.

Heuristic:
delete vertices in order of minimum degree
#Gives <= k+1 colors if all subgraphs have a vertex with degree <=
    k.
```

Theorem 5.4. ***Brooks' Theorem****:*

For a connected graph G *that is neither a complete graph* K_n *nor an odd cycle* C_{2n+1}*:*

$$\chi(G) \leq \Delta(G)$$

6 Intro: Probability

Definition 6.1. *A **discrete probability space** is a set of elementary events often denoted as* $\Omega = \{\omega_1, \omega_2, \ldots, \omega_n\}$.

*Every **elementary event*** ω_i *has a probability* $P[\omega_i] \in [0, 1]$ *such that*

$$\sum_{\omega_i \in \Omega} P[\omega_i] = 1$$

Remark 6.2. *A finite probability space, in which every elementary event has the same probability is called a **Laplace-Space**. Observe that for a Laplace-Space* Ω *and an arbitrary event* E *we have:*

$$P[E] = \frac{|E|}{|\Omega|}$$

Definition 6.3. *An **Event*** $E \subseteq \Omega$ *is a set of elementary events. The probability of an event is defined as:*

$$P[E] = \sum_{\omega_i \in E} P[\omega_i]$$

The complementary event $\bar{E} = \Omega \setminus E$ *has probability:*

$$P[\bar{E}] = 1 - P[E]$$

Theorem 6.4. *For two events* A, B *in a probability space* Ω *we have:*

- $P[\emptyset] = 0$ *and* $P[\Omega] = 1$
- $P[A] \in [0, 1]$
- $P[\bar{A}] = 1 - P[A]$
- *If* $A \subset B$ *then* $P[A] \leq P[B]$

Theorem 6.5. *For events* $A_1, A_2, ...A_n$ *it holds*

$$P[A_1 \cup A_2 \cup \cdots \cup A_n] \leq \sum_{i=1}^{n} P[A_i]$$

with equality iff $A_1, A_2, ...A_n$ *are pairwise disjoint.*

Theorem 6.6. ***Inclusion-Exclusion Principle:*** *For events $A_1, A_2, ...A_n$ with $n \geq 2$ it holds*

$$P[\bigcup_{i=1}^{n} A_i] = \sum_{i=1}^{n} (-1)^{i+1} P[\bigcap_{j=1}^{i} A_j]$$

To proof this theorem we look at an arbitrary $\omega \in \bigcup_{i=1}^{n} A_i$. If ω is in exactly k of the events A_i, then it is counted N_k times where

$$N_k = k - \binom{k}{2} + \binom{k}{3} - \cdots + (-1)^{k+1} \binom{k}{k}$$

We show that $N_k = 1$ for all $k \in \{1, 2, \ldots, n\}$. For that we use the binomial theorem:

$$(x - y)^k = \sum_{i=0}^{k} \binom{k}{i} x^{k-i} (-y)^i$$

Now let $x = 1$ and $y = 1$:

$$(1 - 1)^k = \sum_{i=0}^{k} \binom{k}{i} (-1)^i = \binom{k}{0} - \binom{k}{1} + \binom{k}{2} - \cdots + (-1)^k \binom{k}{k} = 1 - N_k$$

Therfore:

$$N_k = 1 - (1 - 1)^k = 1$$

6.1 Conditional Probability

Definition 6.7. *Let A and B be events with $P[B] > 0$. The* ***conditional probability*** *of A given B is defined as:*

$$P[A \mid B] = \frac{P[A \cap B]}{P[B]}$$

Example 6.8. *Assume a family with two children from which we know that one of them is a girl (B). What is the probability that the other child is also a girl (A)?*

$$P[A \mid B] = \frac{P[A \cap B]}{P[B]} = \frac{1/4}{3/4} = \frac{1}{3}$$

Theorem 6.9. ***Multiplication-Theorem:*** *Let $A_1, A_2, ..A_n$ be events. If $P[A_1 \cap A_2 \cap \cdots \cap A_{n-1}] > 0$ then it holds:*

$$P[A_1 \cap A_2 \cap \cdots \cap A_n] = P[A_1] \cdot P[A_2 \mid A_1] \cdot P[A_3 \mid A_1 \cap A_2] \cdot \cdots \cdot P[A_n \mid A_1 \cap A_2 \cap \cdots \cap A_{n-1}]$$

We can prove the correctness of this theorem by using the definition of conditional probability:

$$P[A_1] \cdot P[A_2 \mid A_1] \cdot P[A_3 \mid A_1 \cap A_2] \ldots P[A_n \mid A_1 \cap A_2 \cap \cdots \cap A_{n-1}]$$

$$= P[A_1] \cdot \frac{P[A_1 \cap A_2]}{P[A_1]} \cdot \frac{P[A_1 \cap A_2 \cap A_3]}{P[A_1 \cap A_2]} \cdots \frac{P[A_1 \cap A_2 \cap \cdots \cap A_n]}{P[A_1 \cap A_2 \cap \cdots \cap A_{n-1}]}$$
$$= P[A_1 \cap A_2 \cap \cdots \cap A_n]$$

Example 6.10. ***Birthday Problem:*** *What is the probability of two people in a room of m having the same birthday? We find the probability that none of them have the same birthday this is $\frac{364}{365} \cdot \frac{363}{365} \cdot \cdots \cdot \frac{365-m+1}{365}$.*

Theorem 6.11. ***Theorem of total probability:*** *Let $A_1, ...A_n$ be pairwise disjoint events and let $B \subseteq A_1 \cup ... \cup A_n$ then it holds:*

$$P[B] = \sum_{i=1}^{n} P[B \mid A_i]P[A_i]$$

Example 6.12. ***Goat Problem:*** *Assume three doors of which we know that one hides a sportscar and the other two a goat. You pick one arbitrary door, then one of the other two opens and you see a goat. Should you now switch to the other door that is left or should you stay with the one you are?*

Let A_1 be the event that you picked the door with the car and A_2 the event that you are infront of the door with a goat. Further let B be the event that you win if you change doors. Now observe tracking-tight

$$Pr[B \mid A_1] = 0$$

$$Pr[B \mid A_2] = 1$$

$$Pr[A_1] = 1/3$$

$$Pr[A_2] = 2/3$$

Using the theorem above we get:

$$Pr[B] = Pr[B \mid A_1]Pr[A_1] + Pr[B \mid A_2]Pr[A_2] = 0 \cdot 1/3 + 1 \cdot 2/3 = 2/3$$

Theorem 6.13. ***Bayes' Theorem:*** *Let $A_1, A_2 ... A_n$ be pariwise disjoint events and $B \subseteq A_1 \cup A_2 ... \cup A_n$ with $P[B] > 0$. Then:*

$$P[A_i \mid B] = \frac{P[B \mid A_i]P[A_i]}{\sum_{j=1}^{n} P[B \mid A_j]P[A_j]}$$

To prove bayes theorem we use the theorem of total probability:

$$P[A_i \mid B] = \frac{P[B \cap A_i]}{P[B]} = \frac{P[B \mid A_i]P[A_i]}{\sum_{j=1}^{n} P[B \mid A_j]P[A_j]}$$

Example 6.14. ***Desease Test:*** *We test a person for a desease that has a probability of $P[A] = 0.01$ to occur. Let A be the event that the tested person has the desease. Further let B be the event that the test was positive. Assume we know the sensitivity of the test $P[B \mid A] = 0.72$ and the specificity $P[\bar{B} \mid \bar{A}] = 0.98$. We want to find the probability that a person has the desease under the condition that the test is positive:*

$$P[A \mid B] = \frac{P[B \mid A]P[A]}{P[B \mid A]P[A] + P[B \mid \bar{A}]P[\bar{A}]} = \frac{0.72 \cdot 0.01}{0.72 \cdot 0.01 + 0.02 \cdot 0.99} = \frac{0.0072}{0.027} \approx 0.267$$

Here we see that even if the test is quite accurate, the probability that a person has the desease is still quite low. This is because the probability of the desease is very low. If the probability of the desease was higher, the probability of the test being positive would be higher.

6.2 Independence

Definition 6.15. *Two events A and B are* ***independent*** *if $P[A \cap B] = P[A]P[B]$ or equivalently $P[A \mid B] = P[A]$.*

Definition 6.16. *Events $A_1, A_2, \ldots, A_n$ are independent if for all subsets $I \subseteq \{1, 2, \ldots, n\}$ it holds:*

$$P[\bigcap_{i \in I} A_i] = \prod_{i \in I} P[A_i]$$

Lemma 6.17. *Let $A_1, A_2 \ldots A_n$ be events and let $A_i^0 := \bar{A}_i$ and $A_i^1 := A_i$ for $i \in \{1, 2, \ldots, n\}$. Then $A_1, A_2 \ldots A_n$ are independent iff for all $(s_1, \ldots s_n) \in \{0, 1\}^n$ it holds that:*

$$P[\bigcap_{i=1}^{n} A_i^{s_i}] = \prod_{i=1}^{n} P[A_i^{s_i}]$$

Lemma 6.18. *Let A, B and C be independent events then so are $A \cap B$ and C and also $A \cup B$ and C.*

7 Random Variables

Definition 7.1. *A **random variable** is a map $X : \Omega \to \mathbb{R}$ where Ω is the set of all events of a probability space.*

Example 7.2. *The event that the random variable X takes a value less than or equal to 5 is denoted as:*

$$X \leq 5 \coloneqq \{\omega \in \Omega \mid X(\omega) \leq 5\}$$

7.1 Expectation

Definition 7.3. *Let X be a random variable. The expectation of X is defined as:*

$$\mathbb{E}[X] \coloneqq \sum_{\alpha \in W_x} \alpha P(X = \alpha)$$

aslong as the sum is absolutely convergent, else we say the expectation is undefined.

Theorem 7.4. *Let X be a random variable with values in $\mathbb{N}_0$. Then:*

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} P(X \geq k)$$

Proof.

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} kP(X = k) = \sum_{k=1}^{\infty} \sum_{j=1}^{k} P(X = k) = \sum_{j=1}^{\infty} \sum_{k=j}^{\infty} P(X = k) = \sum_{j=1}^{\infty} P(X \geq j)$$

□

Example 7.5. *We throw a coin and we know $P[head] = p$. We want to know $\mathbb{E}[X]$ with $X \coloneqq$ number of throws until we get head. Using the above theorem we get:*

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} P(X \geq k) = \sum_{k=1}^{\infty} (1-p)^{k-1} = \frac{1}{1-(1-p)} = \frac{1}{p}$$

Definition 7.6. *For an event $E \subseteq \Omega$ the corresponding indicator-variable X_E is defined as:*

$$X_E(\omega) = \begin{cases} 1 & \text{if } \omega \in E \\ 0 & \text{if } \omega \notin E \end{cases}$$

Remark 7.7. *For an indicator variable X_E we have:*

$$\mathbb{E}[X_E] = P(E)$$

Theorem 7.8. *For random variables $X_1, \dots, X_n$ we have:*

$$X \coloneqq \sum_{i=1}^{n} a_i X_i \mid a_1, \dots, a_n \in \mathbb{R}$$

$$\mathbb{E}[X] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i]$$

This is called the linearity of expectation.

Proof.

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega)P(\omega) = \sum_{\omega \in \Omega} (a_1 X_1(\omega) + \cdots + a_n X_n(\omega))P(\omega)$$

$$= a_1 \sum_{\omega \in \Omega} X_1(\omega)P(\omega) + \cdots + a_n \sum_{\omega \in \Omega} X_n(\omega)P(\omega) = a_1 \mathbb{E}[X_1] + \cdots + a_n \mathbb{E}[X_n]$$

□

Example 7.9. *We throw a coin m times. Let $X \coloneqq$ number of partial sequences with threetimes heads. What is the expectation of X?*

$$X = X_1 + X_2 + \cdots + X_{m-2}$$

where X_i is the indicator variable for the event that the i-th partial sequence is three heads.

$$\mathbb{E}[X_i] = \frac{1}{8}$$

By the linearity of the expectation we get:

$$\mathbb{E}[X] = \sum_{i=1}^{m-2} \mathbb{E}[X_i] = \frac{m-2}{8}$$

Algorithm 7.10

```
#Recall Stable sets in a graph is a set of vertices that are not
    adjacent to each other. We will now use a randomized algorithm
    to determin a maximal stable set.

S[v] = 1/0 with probability p
for all u,v in E:
    if S[u] = 1 and S[v] = 1:
        S[v] = 0
return S
```

this results in a stable set but what is the expected size of S?

Theorem 7.11. *The algorithm produces a stable set of expected size at least* $n * P - m * P^2$

Proof. Let X be the random variable for the size of the stable set S after the first step. Further let Y be the random variable for the amount of edges we looked at in step 2. Observe

$$\mid S \mid \geq X - Y$$

By the linearity of expectation we get:

$$\mathbb{E}[\mid S \mid] \geq \mathbb{E}[X] - \mathbb{E}[Y]$$

We know that $\mathbb{E}[X] = nP$. Further we know that $\mathbb{E}[Y] = mP^2$. Thus we get:

$$\mathbb{E}[\mid S \mid] \geq nP - mP^2$$

□

7.2 Variance

Definition 7.12. *For a random variable* X *with* $\mathbb{E}[X] = \mu$ *the variance is defined as:*

$$Var[X] := \mathbb{E}[(X - \mu)^2]$$

The standard deviation is defined as $\sigma := \sqrt{Var[X]}$.

Theorem 7.13. *For a random variable* X *we have* $Var[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$

Lemma 7.14. *For a random variable* X *and* $a, b \in \mathbb{R}$ *we have* $Var[aX + b] = a^2 Var[X]$

8 Distributions

Definition 8.1. *Let X be a random variable with values in $\{0, 1\}$ and density*

$$P[X = 1] = p \quad \text{and} \quad P[X = 0] = 1 - p$$

Then X follows a Bernoulli distribution with parameter p, denoted as $X \sim \text{Bernoulli}(p)$.

Definition 8.2. *Let X be a random variable with values in $\{0, 1, \ldots, n\}$ and density*

$$P[X = k] = \binom{n}{k} p^k (1 - p)^{n-k}$$

Then X follows a Binomial distribution with parameters n and p, denoted as $X \sim \text{Bin}(n, p)$.

Definition 8.3. *Let X be a random variable with density*

$$P[X = k] = \frac{e^{-\lambda} \lambda^k}{k!}$$

Then X follows a Poisson distribution with parameter λ, denoted as $X \sim \text{Po}(\lambda)$.

Definition 8.4. *Let X be a random variable with density*

$$P[X = k] = p(1 - p)^{k-1}$$

Then X follows a Geometric distribution with parameter p, denoted as $X \sim \text{Geo}(p)$.

Example 8.5. *Assume we throw a coin with $P[\text{head}] = p$ until we get n times heads. Let X be the number of throws until we get n times heads.*

$$P[X = k] = \binom{k-1}{n-1} p^n (1 - p)^{k-n}$$

Definition 8.6. *For $n \geq 2$ a random variable X with density*

$$P[X = k] = \binom{k-1}{n-1} p^n (1 - p)^{k-n}$$

is called negative binomial distributed with order n.

Example 8.7. *There are n diffrent types of collectible cards. In every round we uniformly get one of them. Let X be the number of rounds until we have all n types of cards. What is $\mathbb{E}[X]$?*